



Lecture 7: State-space form, Linearization

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Disclaimer: These (informal) notes are not intended to replace the textbook or the material contained therein that students are expected to cover and study. If you see any typos, please let me know.

1 Outline

In the previous lecture..

- Sensitivity and robustness

Today:

- State-space representation
- Linearization

Today, we change our mathematical representation, moving from the frequency domain to the time domain and from classical to modern control. We move from Laplace transforms and transfer functions to the state-space representation. We will conclude by looking at how a general nonlinear differential equation can be approximated by a linear differential equation.

Table 1: Today's Ordlista

English	Svenska
state-space (form/representation)	tillstånds(form/beskrivning)
linearization	linjärisering
state vector	tillståndsvektorn
Diagonal form	Diagonalform
Controllable canonical form	Styrbar kanonisk form
Observable canonical form	Observerbar kanonisk form
Fixed point	Stationär punkt
eigenvalues	egenvärden
Nonlinear	Olinjärt

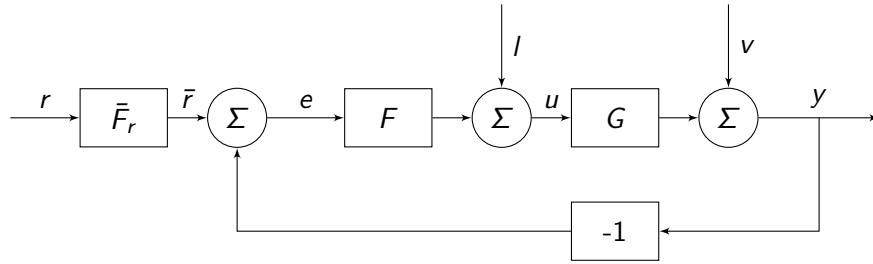


Figure 1: General closed-Loop block diagram. It should not be surprising that $1 + GF$ is present in all denominators.

2 Recap

2.1 General Closed-Loop Structure

Consider a feedback control system comprising a controller F , a plant G , a reference pre-filter $\bar{F}_r$, an input disturbance l , and an output disturbance v (Figure 1). The output and the control signal are, respectively,

$$Y = \underbrace{\frac{GF}{1+GF}}_{G_c} \bar{R} + \underbrace{\frac{1}{1+GF}}_S V + \underbrace{\frac{G}{1+GF}}_{G_{ly}} L,$$

$$U = \underbrace{\frac{F}{1+GF}}_{G_{ru}} (\bar{R} - V) + \underbrace{\frac{1}{1+GF}}_S L.$$

The central task of control design is to determine F (equivalently, the open-loop transfer function $G_o = GF$) such that the closed-loop system meets desired performance and robustness specifications.

2.2 Four Performance Tests

A systematic way to evaluate the quality of a controller design consists of plotting the magnitudes of four key transfer functions as a function of frequency:

1. **Closed-loop transfer function** $|G_c(i\omega)|$. The bandwidth ω_B and the resonance peak M_p characterize servo performance. The design goal is $G_c(i\omega) \approx 1$ for $\omega < \omega_B$, i.e., near-perfect reference tracking within the system bandwidth. Check Fig. 5.8 in the book.
2. **Sensitivity function** $|S(i\omega)|$. Since $S(i\omega) + G_c(i\omega) = 1$, one has $S(i\omega) \approx 0$ for $\omega < \omega_B$. This quantifies how well output disturbances are attenuated.
3. **Input-disturbance transfer function** $|G_{ly}(i\omega)|$. One can show that $G_{ly} = \frac{G}{1+GF} \frac{F}{F} = G_c \frac{1}{F} \approx 1/F$ for $\omega < \omega_B$, so a high-gain controller is beneficial for the rejection of input disturbances and measurement noise.
4. **Control-effort transfer function** $|G_{ru}(i\omega)|$. One has $G_{ru} = \frac{F}{1+GF} \frac{G}{G} = G_c \frac{1}{G} \approx 1/G$ for $\omega < \omega_B$, which provides information about the magnitude of the control signal u (important whenever actuator saturation limits are present).

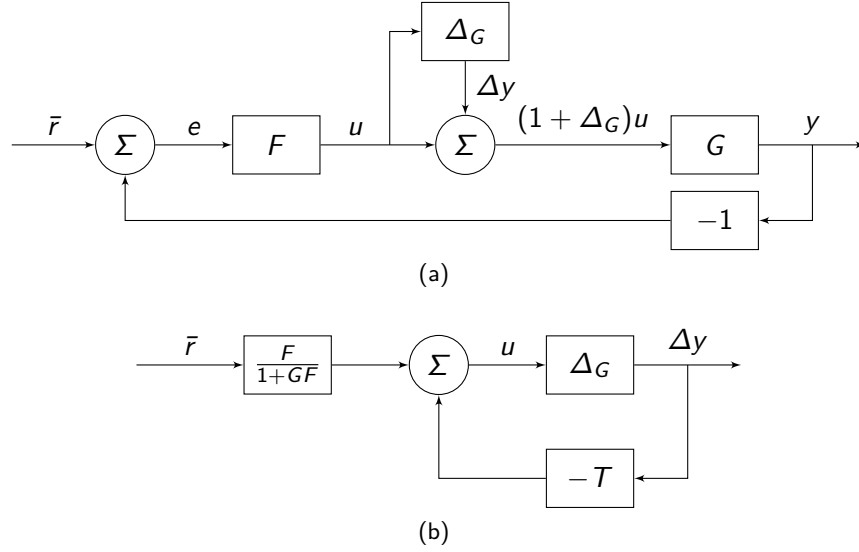


Figure 2: Block Diagram: Robustness Analysis.

2.3 Robustness Analysis

Suppose the true plant is $G^0(s) = G(s)(1 + \Delta_G(s))$, where Δ_G represents the model relative error, see Figure 2a. From Figure 2a, we derive:

$$\begin{cases} U &= -\frac{GF}{1+GF} \Delta Y + \frac{F}{1+GF} \bar{R} \\ \Delta Y &= \Delta_G U \\ T &= G_c = \frac{GF}{1+GF} \end{cases}$$

Remark 2.1. Observe that the transfer function from $\bar{r}$ to u is FS , and the transfer function from Δ_y to u is T , the complementary sensitivity function that we introduced in F6, which is equivalent to G_c (using $F_y = F$ and $F_r = \bar{F}_r F$ in the notation introduced in F6).

Focusing on ΔY , it is useful to redraw this relationship in a new block diagram as presented in Figure 2b. The standard robustness criterion (see F6) states that the closed-loop system is stable for all admissible perturbations if and only if

$$|T(i\omega)| < \frac{1}{|\Delta_G(i\omega)|} \quad \forall \omega, \quad (1)$$

where $T = \frac{GF}{1+GF} = G_c$ is the complementary sensitivity function. This criterion follows from a Nyquist argument applied to the feedback interconnection of T and Δ_G (Figure 2b): the Nyquist curve of $T(i\omega)\Delta_G(i\omega)$ must not encircle -1 , which is guaranteed (as a sufficient condition) whenever $|T(i\omega)\Delta_G(i\omega)| < 1$ for all ω .

2.4 Reference Pre-Filter F_r

Adding a reference pre-filter F_r so that $\bar{r} = F_r r$, does *not* affect the sensitivity function S or the robustness margin, since F_r does not enter the feedback loop. Its sole effect is on the servo response $r \rightarrow y$. A common choice is a low-pass filter, which smooths step references and thereby limits the peak magnitude of the control signal u , potentially preventing actuator saturation.

3 State-space Representation

3.1 Motivation and “Limitations” of Transfer Functions

Throughout the course, dynamical systems have been described by differential equations or, equivalently in the Laplace domain, by transfer functions.

$$\ddot{y}(t) + a_1\dot{y}(t) + a_2y(t) = b_0\dot{u}(t) + b_1u(t) \Rightarrow G(s) = \frac{b_0s + b_1}{s^2 + a_1s + a_2} \quad (2)$$

While transfer functions are powerful tools for frequency-domain analysis and controller design, it may also be easier to miss other important information, such as *nonzero initial conditions*. This can lead to subtly incorrect conclusions, as the following example illustrates.

Example 1

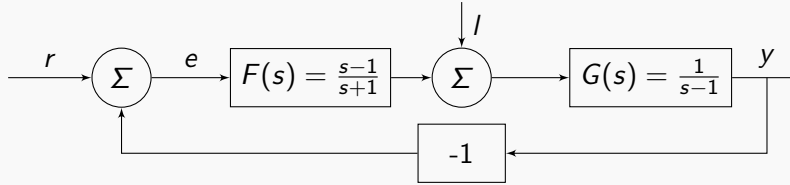
Consider the unstable plant $G(s) = \frac{1}{s-1}$ and the controller $F(s) = \frac{s-1}{s+1}$. The closed-loop transfer function from reference to output is

$$G_c = \frac{GF}{1 + GF} = \frac{1}{s+2} \quad (\text{stable}),$$

while the transfer function from the input disturbance I to the output is

$$G_{I/y} = \frac{G}{1 + GF} = \frac{s+1}{(s+2)(s-1)} \quad (\text{unstable!}).$$

The algebraic cancellation of the factor $(s-1)$ in G_c masks a hidden unstable pole. In the presence of non-zero initial conditions or disturbances the system will exhibit unstable behavior. The key lesson is that unstable right-half-plane poles must never be canceled by controller zeros, as this hides rather than stabilizes the unstable mode.



3.2 The State-Space Model

The state-space representation overcomes the limitation described above by working directly in the time domain and tracking the full internal state of the system. A general n -th order ($n \geq 2$) linear time-invariant (LTI) system is written as

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0, \quad (3)$$

$$y(t) = Cx(t) + Du(t), \quad (4)$$

where $x(t) \in \mathbb{R}^n$ is the *state vector*, $u(t) \in \mathbb{R}^m$ the input, $y(t) \in \mathbb{R}^p$ the output, and $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$, $D \in \mathbb{R}^{p \times m}$ are constant matrices. In most physical applications, $D = 0$.

The state vector $x(t) = [x_1(t) \ \cdots \ x_n(t)]^T$ encapsulates all internal information required to determine the future evolution of the system given any future input. The initial condition $x(0) = x_0$ (for $t = 0$) explicitly accounts for the system's history.

3.3 From Differential Equations to State-Space Models

Several systematic approaches exist to construct a state-space realization from a differential equation or transfer function.

- I. Physical states [Exempel 8.2, 8.3]
- II. Diagonal form [Exempel 8.5]
- III. Controllable canonical form [Exempel 8.6]
- IV. Observable canonical form [Exempel 8.7]

Example 2: I. Physical states

Consider a point mass m subject to an applied force F (the control input u), with position y as the output. Newton's second law gives $m\ddot{y} = F$, a second-order equation known as the double integrator. Defining $x_1(t) = y(t)$ (position) and $x_2(t) = \dot{y}(t)$ (velocity), the state-space model is

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} u, \\ y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{cases}$$

This approach is the most natural whenever state variables admit direct physical interpretations.

Example 3: II. Diagonal form (via partial fractions)

For a DC motor with transfer function $G(s) = \frac{1}{s(s+1)}$, a partial-fraction expansion yields

$$G(s) = \frac{1}{s(s+1)} = \frac{1}{s} + \frac{-1}{s+1}$$

and the output signal is then obtained by

$$Y(s) = \underbrace{\frac{1}{s}U(s)}_{X_1(s)} + \underbrace{\frac{-1}{s+1}U(s)}_{X_2(s)}.$$

Identifying $X_1(s) = \frac{1}{s}U(s)$ and $X_2(s) = \frac{-1}{s+1}U(s)$, that is, $sX_1 = U$ and $(s+1)X_2 = -U$ means $sx_2 = -x_2 - U$, one obtains

$$\dot{x}_1 = u, \quad \dot{x}_2 = -x_2 - u, \quad y = x_1 + x_2,$$

corresponding to

$$\begin{cases} \dot{x}_1(t) = u(t) \\ \dot{x}_2(t) = -x_2(t) - u(t) \\ y(t) = x_1(t) + x_2(t) \end{cases} \Leftrightarrow A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad C = [1 \quad 1], \quad D = 0.$$

The diagonal structure of A reflects the decoupled nature of the modes, each corresponding to a pole of the transfer function.

3.4 From State-Space Form to Transfer Function

Applying the Laplace transform to (4) under the assumption $x(0) = 0$ gives

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases} \xrightarrow{\mathcal{L}} \begin{cases} sX(s) = AX(s) + BU(s) \\ Y(s) = CX(s) + DU(s) \end{cases}$$

That is:

$$(sI - A)X(s) = BU(s) \Rightarrow X(s) = (sI - A)^{-1}BU(s)$$

and

$$Y(s) = CX(s) + DU(s) = \left(C(sI - A)^{-1}B + D \right) U(s)$$

This yields

$$\boxed{G(s) = D + C(sI - A)^{-1}B} \quad (5)$$

Check Resultat 8.4 in the book.

Remark 3.1. Remember your linear algebra course to compute the inverse of a matrix. For example, the rule for the inverse of 2×2 matrices is:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow M^{-1} = \frac{1}{\det M} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

That is: (1) Change the place of diagonal elements, (2) swap the off-diagonal elements with a $-$ sign, and (3) divide by the determinant.

Example 4: Double integrator

Given the system

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x\end{aligned}$$

one computes

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 0 & s \end{bmatrix} \Rightarrow (sI - A)^{-1} = \frac{1}{s^2} \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix}$$

yielding $G(s) = \frac{1}{s^2}$, i.e., the double-integrator ($\ddot{y} = u$) transfer function.

3.5 Poles and Eigenvalues

The *poles of the system* are the values of s for which $(sI - A)$ is singular, i.e., the roots of the *characteristic polynomial* $\det(sI - A)$. These are precisely the eigenvalues of A :

$$Av = \lambda v \Rightarrow (\lambda I - A)v = 0 \Rightarrow \det(\lambda I - A) = 0.$$

More generally, writing $(sI - A)^{-1} = \text{adj}(sI - A) / \det(sI - A)$, where $(sI - A)^*$ denotes the adjoint matrix, equation (5) takes the form

$$G(s) = C \frac{(sI - A)^*}{\det(sI - A)} B + D,$$

establishing the fundamental result (Resultat 8.4 in the book):

$$\boxed{\text{poles of the system} = \text{eigenvalues of } A.} \quad (6)$$

3.6 Stability

Definition 3.2 (Asymptotic stability). The system $\dot{x}(t) = Ax(t)$ (with $x(0) = x_0$) is *asymptotically stable* if $\lim_{t \rightarrow \infty} x(t) = 0$ for all initial conditions $x_0 \in \mathbb{R}^n$.

Theorem 3.3 (Resultat 8.7). A linear system is asymptotically stable if and only if all eigenvalues of A have strictly negative real parts, i.e., lie in the open left-half plane (LHP).

Theorem 3.4 (Resultat 8.8). A linear system is input-output (I/O) stable if all eigenvalues of A lie in the open LHP.

These two theorems are consistent with the classical stability criterion for transfer functions (compare Resultat 8.8 and Sats 2.2), confirming that the state-space and frequency-domain frameworks are compatible.

4 Linearization of Nonlinear Systems

4.1 Nonlinear State-Space Models

Many physical systems of practical interest are inherently nonlinear. Their dynamics are described by

$$\begin{cases} \dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= h(x(t), u(t)) \end{cases} \quad (7)$$

where $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $h : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^p$ are smooth (generally nonlinear) functions. An example is

$$\begin{cases} \dot{x} &= x + x^2 + \sin u \\ y &= e^x + \cos u \end{cases}$$

The linear analysis tools developed in the preceding lectures (transfer functions, frequency response, etc.) cannot be applied directly to systems of the form (7). However, under appropriate conditions, a useful linear approximation can be derived.

4.2 Equilibrium Points

Assume the system is driven by a constant input $u(t) = \bar{u}$. If the state converges to a constant value $\bar{x}$ as $t \rightarrow \infty$, then $\dot{x}(t) \rightarrow 0$, and the triple $(\bar{x}, \bar{u}, \bar{y})$ is called a *equilibrium point* (or fixed point, stationary point) of the system, satisfying

$$f(\bar{x}, \bar{u}) = 0, \quad h(\bar{x}, \bar{u}) = \bar{y}. \quad (8)$$

A water-tank system provides a concrete physical example: under a constant inflow, the water level eventually reaches a constant equilibrium $\bar{x}$ corresponding to a balance between inflows and outflows.

4.3 Linearization via Taylor Expansion

To analyze the behavior of the system near an equilibrium point, consider small perturbations (variations around this point):

$$\Delta x = x - \bar{x}, \quad \Delta u = u - \bar{u}, \quad \Delta y = y - \bar{y}.$$

Expanding f in a Taylor series around $(\bar{x}, \bar{u})$, the derivative of Δx can be calculated as:

$$\begin{aligned} \Delta \dot{x} &= \dot{x} = f(\bar{x} + \Delta x, \bar{u} + \Delta u) \\ &= \underbrace{f(\bar{x}, \bar{u})}_{=0} + f_x(\bar{x}, \bar{u}) \Delta x + f_u(\bar{x}, \bar{u}) \Delta u + \text{higher-order terms (h.o.t.)}, \end{aligned}$$

where $f(\bar{x}, \bar{u}) = 0$ by the equilibrium-point condition (8) and $\frac{d}{dt}\bar{x} = 0$ since $\bar{x}$ is constant. Applying the same procedure to the output equation and neglecting higher-order terms (valid for sufficiently small perturbations),

$$\begin{aligned} \Delta y &= y - \bar{y} = h(\bar{x} + \Delta x, \bar{u} + \Delta u) - \bar{y} = \{ \text{Taylor} \} = \\ &= \underbrace{h(\bar{x}, \bar{u}) - \bar{y}}_{=0} + h_x(\bar{x}, \bar{u}) \Delta x + h_u(\bar{x}, \bar{u}) \Delta u + \text{h.o.t.}, \end{aligned}$$

one obtains the *linearized system* (Resultat 8.1):

$$\begin{aligned} \Delta \dot{x} &= A \Delta x + B \Delta u, \\ \Delta y &= C \Delta x + D \Delta u, \end{aligned} \quad (9)$$

where the matrices A, B, C, D are the **Jacobians** of f and h evaluated at the fixed point:

$$\begin{aligned} A &= f_x(\bar{x}, \bar{u}) = \left. \frac{\partial f}{\partial x} \right|_{(\bar{x}, \bar{u})}, & B &= f_u(\bar{x}, \bar{u}) = \left. \frac{\partial f}{\partial u} \right|_{(\bar{x}, \bar{u})}, \\ C &= h_x(\bar{x}, \bar{u}) = \left. \frac{\partial h}{\partial x} \right|_{(\bar{x}, \bar{u})}, & D &= h_u(\bar{x}, \bar{u}) = \left. \frac{\partial h}{\partial u} \right|_{(\bar{x}, \bar{u})}. \end{aligned} \quad (10)$$

For example, the Jacobian matrix $A = f_x$ can be calculated as follows:

$$f(x, u) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n, u) \\ \vdots \\ f_n(x_1, x_2, \dots, x_n, u) \end{bmatrix} \Rightarrow A = \begin{bmatrix} \nabla f_1 \\ \vdots \\ \nabla f_n \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}_{(\bar{x}, \bar{u})}.$$

Remark 4.1. The linear approximation (9) is valid only for *small* deviations Δx , Δu , and Δy from the equilibrium point. For large perturbations, the neglected higher-order terms become significant and the linear model loses validity.

4.4 Local Stability via Linearization

A fundamental question is whether the stability of the linearized model implies stability of the original nonlinear system near the equilibrium point.

Theorem 4.2 (Lyapunov's Indirect Method). *If all eigenvalues of the Jacobian matrix $A = f_x(\bar{x}, \bar{u})$ defined in (10) lie strictly in the open left-half plane, then the nonlinear system (7) is locally asymptotically stable in a neighborhood of the equilibrium $(\bar{x}, \bar{u})$.*

The formal proof of Theorem 4.2 relies on **Grönwall's inequality**, attributed to the Swedish-born mathematician Thomas Grönwall:

Lemma 4.3 (Grönwall's inequality). *If $f(t)$ is continuous and*

$$f(t) \leq \alpha + \beta \int_0^t f(s) ds, \quad \forall t > 0,$$

with $\alpha \leq 0$ and $\beta \leq 0$, then $f(t) \leq \alpha e^{\beta t}$ for all $t > 0$.

Applied to the error dynamics between the solutions of the nonlinear and linearized systems, this inequality establishes that small initial deviations decay exponentially to zero, provided the linearized system is asymptotically stable.

Remark 4.4. Theorem 4.2 is a *local* result: it guarantees stability only in a neighborhood of the fixed point. The extent of this neighborhood, i.e., the *region of attraction*, cannot be determined from the linear analysis alone and requires separate Lyapunov-function arguments.